MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 05
Chapter 10: Circles

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. A right triangle ABC with $\angle B = 90°$, $AB = 6$ cm, and $BC = 8$ cm circumscribes a circle. The inradius r of the circle is: [1]

(A) 1 cm (B) 2 cm (C) 3 cm (D) 4 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 2 cm
Justification: Hypotenuse $AC = \sqrt{6^2 + 8^2} = 10$ cm. Inradius $r = \frac{AB+BC-AC}{2} = \frac{6+8-10}{2} = \boxed{\mathbf{2}\text{ cm}}$.

Q2. If PA and PB are two tangents drawn from an external point P to a circle of radius 4 cm such that $PA \perp PB$, then the length of chord AB is: [1]

(A) 4 cm (B) $4\sqrt{2}$ cm (C) 8 cm (D) $2\sqrt{2}$ cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) $4\sqrt{2}$ cm
Justification: Since $PA \perp PB$, $OAPB$ is a square of side $r = 4$ cm. Chord AB is the diagonal of this square, so $AB = r\sqrt{2} = \boxed{\mathbf{4\sqrt{2}}\text{ cm}}$.

Q3. If radii of two concentric circles are 17 cm and 8 cm, the length of the chord of the larger circle tangent to the smaller circle is: [1]

(A) 15 cm (B) 25 cm (C) 30 cm (D) 32 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) 30 cm
Justification: Half-chord $= \sqrt{17^2 - 8^2} = \sqrt{225} = 15$ cm $\implies$ Chord $= 2 \times 15 = \boxed{\mathbf{30}\text{ cm}}$.

Q4. From an external point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is: [1]

(A) 7 cm (B) 12 cm (C) 15 cm (D) 24.5 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 7 cm

Justification: Radius $r = \sqrt{OQ^2 - \text{tangent}^2} = \sqrt{25^2 - 24^2} = \sqrt{49} = \boxed{\mathbf{7}\text{ cm}}$.

Q5. **Assertion (A):** If the angle between two tangents drawn from an external point P to a circle with centre O is $60°$, then $\angle AOB = 120°$.
Reason (R): The angle between two tangents drawn from an external point is supplementary to the angle subtended by the line segments joining the points of contact at the centre. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

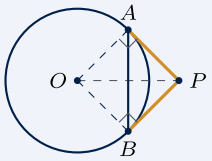
Justification: By Reason (R), $\angle APB + \angle AOB = 180° \implies 60° + \angle AOB = 180° \implies \angle AOB = 120°$. Thus both are true and (R) explains (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. If PA and PB are tangents from external point P to a circle of radius 5 cm with centre O such that $PA \perp PB$, find the length of the chord AB. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

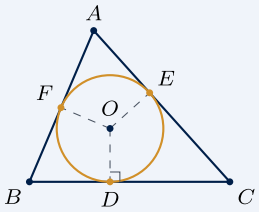
Step 1: Identifying Quadrilateral $OAPB$ [1 Mark]
Since $\angle OAP = \angle OBP = 90°$, $\angle APB = 90°$, and $OA = OB = 5$ cm, $OAPB$ is a square of side 5 cm ($PA = PB = 5$ cm).



Step 2: Diagonal / Chord AB [1 Mark]
In right $\triangle APB$: $AB = \sqrt{PA^2 + PB^2} = \sqrt{5^2 + 5^2} = \boxed{\mathbf{5\sqrt{2}}\text{ cm}}$.

Q7. In $\triangle ABC$, an incircle touches sides BC, CA, AB at D, E, F respectively. If $AB = c, BC = a, CA = b$ and s is the semi-perimeter, prove that $AF = s - a$, $BD = s - b$, and $CE = s - c$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Let $AF = AE = x$, $BD = BF = y$, and $CE = CD = z$ (by Theorem 10.2). [1 Mark]

Step 1: Side Lengths & Semi-Perimeter [1 Mark]

$x + y = c, \quad y + z = a, \quad z + x = b.$

Adding gives $2(x + y + z) = a + b + c = 2s \implies x + y + z = s.$

Step 2: Solving for x, y, z [1 Mark]

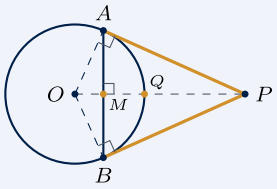
$AF = x = s - (y + z) = s - a; \quad BD = y = s - (x + z) = s - b;$

$CE = z = s - (x + y) = s - c.$

$$\boxed{\therefore AF = s - a, \quad BD = s - b, \quad CE = s - c \quad \text{[Proved]}}$$

Q8. Tangents PA and PB are drawn to a circle with centre O from an external point P. If OP intersects the circle at Q and chord AB at M, prove that $PM \cdot PO = PA^2$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Perpendicularity of OP on Chord AB [1 Mark]

$\triangle OAP \cong \triangle OBP \implies \angle APO = \angle BPO$. In isosceles $\triangle PAB$, bisector PM of vertex $\angle APB$ is perpendicular to base AB, so $\angle PMA = 90°$.

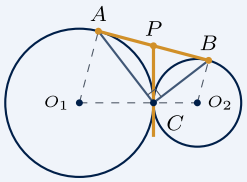
Step 2: Similarity $\triangle PMA \sim \triangle PAO$ [2 Marks]

In $\triangle PMA$ and $\triangle PAO$: $\angle PMA = \angle PAO = 90°$ and $\angle MPA = \angle APO$ (common). By AA similarity, $\triangle PMA \sim \triangle PAO$:

$$\frac{PM}{PA} = \frac{PA}{PO} \implies \boxed{PM \cdot PO = PA^2 \quad \text{[Proved]}}$$

Q9. Two circles touch each other externally at point C. A common tangent AB touches the circles at A and B. Prove that the common tangent at C bisects AB, and that $\angle ACB = 90°$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Let the common tangent at C meet direct common tangent AB at P.

Step 1: Common Tangent Bisects AB [1.5 Marks]

From point P: $PA = PC$ (tangents to left circle) and $PB = PC$ (tangents to right circle). Hence $PA = PB$, i.e., P bisects AB.

Step 2: Proving $\angle ACB = 90°$ [1.5 Marks]

In $\triangle PAC$, $PA = PC \implies \angle PAC = \angle PCA$. In $\triangle PBC$, $PB = PC \implies \angle PBC = \angle PCB$.

In $\triangle ABC$, $\angle PAC + \angle PBC + \angle ACB = 180° \implies 2\angle ACB = 180°$.

$$\boxed{\therefore PA = PB \quad \text{and} \quad \angle ACB = 90° \quad \text{[Proved]}}$$

Q10. Architectural Arch and Supporting Tangent Struts

An architect designs a semicircular decorative glass dome of radius $r = 4$ m resting on diameter AB with centre O. Two external support struts TP and TQ touch the circular boundary at points P and Q and meet at apex T. The struts are inclined to each other at $\angle PTQ = 90°$.

(i) Classify the geometric shape of quadrilateral $OPTQ$. [1]

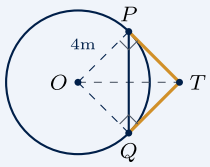
(ii) Find the length of apex strut TP. [1]

(iii) Calculate the direct line-of-sight distance OT from centre to apex, OR find the length of chord PQ. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Classification of $OPTQ$: [1 Mark]

$\angle OPT = \angle OQT = \angle PTQ = 90° \implies \angle POQ = 90°$, and adjacent sides $OP = OQ = 4$ m. Hence $OPTQ$ is a $\boxed{\textbf{Square}}$.



(ii) Length of Apex Strut TP: [1 Mark]

All sides of square $OPTQ$ are equal: $TP = OP = \boxed{\textbf{4 m}}$.

(iii) Main Choice — Direct Distance OT: [2 Marks]

OT is the diagonal of square $OPTQ$ (in right $\triangle OPT$): $OT = \sqrt{OP^2 + TP^2} = \sqrt{4^2 + 4^2} = \boxed{\textbf{4}\sqrt{\textbf{2}}\ \text{m} \approx \textbf{5.66}\ \text{m}}$.

— OR — (Alternative Choice — Length of Chord PQ): [2 Marks]

PQ is the other diagonal of square $OPTQ$ (in right $\triangle PTQ$):

$$\boxed{PQ = \sqrt{TP^2 + TQ^2} = \sqrt{4^2 + 4^2} = \textbf{4}\sqrt{\textbf{2}}\ \text{m} \quad (\approx \textbf{5.66}\ \text{m})}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★